Student Club Registration Management

Category B Practice - Part A, Problem 3

Problem summary

Members are unique by roll number only, even if incoming Student instances differ by name. Override equals() and hashCode() using rollNumber alone.

Java solution

```
import java.util.*;
public class Main {
 static class Student {final String roll,name;Student(String r,String n){roll=r;name=n;}
  public boolean equals(Object o){if(this==o)return true;if(!(o instanceof Student))return false;return roll.equ
  public int hashCode(){return roll.hashCode();}}
 public static void main(String[] args){Scanner sc=new Scanner(System.in);Set<Student> members=new HashSet<>();L
  while(sc.hasNextLine()){String line=sc.nextLine().trim();if(line.isEmpty())continue;String[] p=line.split("\\s
   if(p[0].equals("ADD")&&p.length==3)System.out.println(members.add(new Student(p[1],p[2]))?"Added":"duplicate
   else if(p[0].equals("CONTAINS")&&p.length==3)checks.add("contains: "+members.contains(new Student(p[1],p[2]))
  }
  System.out.println("member count "+members.size());for(String c:checks)System.out.println(c);
 }
}
```

Example input

```
ADD 21CS01 Asha
ADD 21CS01 Asha
ADD 21CS02 Ravi
CONTAINS 21CS02 Ravi
CONTAINS 21CS03 Priya
```

Expected output

```
Added
duplicate rejected
Added
member count 2
contains: true
contains: false
```

Notes

Average HashSet add/contains time is O(1). Both equality methods use roll number only, as required.